

COMS3003A Tutorial 8

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Cipher:

6b7662747a7374787a6d78717a766167776a747763777a7679756269746f6869756f6c6d63647169
6c76776268706975726f757177

Hint: Cumberbatch

1. Determine if the following formula are satisfied by the assignment:

1. $v(A) = T, v(B) = T, v(C) = T, (A \rightarrow B) \rightarrow C$
2. $v(A) = T, v(B) = F, v(C) = T, (A \rightarrow B) \rightarrow C$
3. $v(A) = T, v(B) = F, v(C) = F, ((A \wedge B) \rightarrow (B \wedge C)) \vee \neg(A \wedge \neg B \wedge C)$
4. $v(A) = T, v(B) = T, v(C) = T, ((A \wedge B) \rightarrow (B \wedge C)) \vee \neg(A \wedge \neg B \wedge C)$

1. $v(A) = T, v(B) = T, v(C) = T, (A \rightarrow B) \rightarrow C$

Let $v \models A$ and $v \models B$ and $v \models C$

$v \models A$ and $v \models B$

$\Leftrightarrow v \models A \wedge B$

$\Leftrightarrow v \models A \rightarrow B$

$v \models A \rightarrow B$ and $v \models C$

$\Leftrightarrow v \models (A \rightarrow B) \wedge C$

$\Leftrightarrow v \models (A \rightarrow B) \rightarrow C$

2. $v(A) = T, v(B) = F, v(C) = T, (A \rightarrow B) \rightarrow C$

Let $v \models A$ and $v \not\models B$ and $v \models C$

$v \models A$ and $v \not\models B$

$\Leftrightarrow v \not\models A \rightarrow B$

$v \not\models A \rightarrow B$

$\Leftrightarrow v \models (A \rightarrow B) \rightarrow C$ vacuously

Technically speaking, we do not need to specify that $v \models (A \rightarrow B) \rightarrow C$ vacuously.

I am leaving it in here so you can identify what definition is being applied. In your own answers, you may leave it out.

I would still encourage you leave it in as it can help simplify your thought process

3. $v(A) = T, v(B) = F, v(C) = F, ((A \wedge B) \rightarrow (B \wedge C)) \vee \neg(A \wedge \neg B \wedge C)$

Let $v \models A$ and $v \not\models B$ and $v \not\models C$

$v \models A$ and $v \not\models B$

$\Leftrightarrow v \not\models (A \wedge B)$

$\Leftrightarrow v \models (A \wedge B) \rightarrow (B \wedge C)$ vacuously

$\Leftrightarrow v \models ((A \wedge B) \rightarrow (B \wedge C)) \vee \neg(A \wedge \neg B \wedge C)$

We can go down either branch of an or statement to show it is true.

We also show that the implication $(A \wedge B) \rightarrow (B \wedge C)$ is vacuously true. Vacuous truth is incredibly useful like this, we don't even need to care about $B \wedge C$ until we have first shown that the antecedent (first part of the implication $A \wedge B$) is true.

In this particular assignment, as $A \wedge B$ is not true, we can immediately skip over the formula.

This is the exact same process computers use to *short-circuit* the evaluation of a boolean expression. *Don't evaluate the entire expression until it is absolutely necessary.*

4. $v(A) = T, v(B) = T, v(C) = T, ((A \wedge B) \rightarrow (B \wedge C)) \vee \neg(A \wedge \neg B \wedge C)$

Let $v \models A$ and $v \not\models B$ and $v \not\models C$

2. Prove (or disprove) the following formula are satisfiable:

1. $A \wedge \neg A$
2. $A \vee \neg A$
3. $A \wedge (B \vee \neg A)$
4. $\neg A \wedge (A \rightarrow B)$

1. $A \wedge \neg A$

Assume $A \wedge \neg A$ is satisfiable

$\Leftrightarrow$ There is some v such that $v \models A \wedge \neg A$

$\Leftrightarrow v \models A$ and $v \models \neg A$

$\Leftrightarrow v \models A$ and $v \not\models A$

Therefore, by contradiction, $A \wedge \neg A$ is not satisfiable.

2. $A \vee \neg A$

Let $v \models A$

$\Leftrightarrow v \models A$ or anything that is true or false

$\Leftrightarrow v \models A$ or $v \models \neg A$

$\Leftrightarrow v \models A \vee \neg A$

The line where we say $v \models A$ or anything is called Disjunctive Introduction

Therefore, $A \vee \neg A$ is satisfiable.

3. $A \wedge (B \vee \neg A)$

Let $v \models A$ and $v \models B$
 $\Rightarrow v \models B \vee \neg A$
 $\Rightarrow v \models A \wedge (B \vee \neg A)$

Therefore $A \wedge (B \vee \neg A)$ is satisfiable

4. $\neg A \wedge (A \rightarrow B)$

Let $v \not\models A$
 $\Rightarrow v \models \neg A$ and $v \models A \rightarrow B$ vacuously
 $\Rightarrow v \models \neg A \wedge (A \rightarrow B)$

Therefore $\neg A \wedge (A \rightarrow B)$ is satisfiable.

3. Prove (or disprove) the following formula are valid:

- You may *not* assume that $\models \varphi \iff \neg \varphi$ is not satisfiable

- $(A \wedge (A \rightarrow B)) \rightarrow B$
 - $(\neg B \wedge (A \rightarrow B)) \rightarrow \neg A$
 - $(A \rightarrow B \wedge B \rightarrow C) \rightarrow (A \rightarrow C)$
 - $(A \rightarrow B) \leftrightarrow (\neg B \rightarrow \neg A)$
-

1. $(A \wedge (A \rightarrow B)) \rightarrow B$

Assume $(A \wedge (A \rightarrow B)) \rightarrow B$ is not valid
 $\Rightarrow$ There is some v such that $v \not\models (A \wedge (A \rightarrow B)) \rightarrow B$
 $\Rightarrow v \models A \wedge (A \rightarrow B)$ and $v \not\models B$
 $\Rightarrow v \models A$ and $v \models (A \rightarrow B)$ and $v \not\models B$
 $\Rightarrow v \models B$ as $v \models A$ and $v \models A \rightarrow B$
 $\Rightarrow v \models B$ and $v \not\models B$

Therefore by contradiction, $(A \wedge (A \rightarrow B)) \rightarrow B$ is valid.

2. $(\neg B \wedge (A \rightarrow B)) \rightarrow \neg A$

Assume $(\neg B \wedge (A \rightarrow B)) \rightarrow \neg A$ is not valid
 $\Rightarrow$ There is some v such that $v \not\models (\neg B \wedge (A \rightarrow B)) \rightarrow \neg A$
 $\Rightarrow v \models \neg B \wedge (A \rightarrow B)$ and $v \not\models \neg A$
 $\Rightarrow v \models \neg B$ and $v \models A \rightarrow B$ and $v \models A$
 $\Rightarrow v \models B$ as $v \models A \rightarrow B$ and $v \models A$
 $\Rightarrow v \models B$ and $v \models \neg B$

Therefore by contradiction, $(A \wedge (A \rightarrow B)) \rightarrow B$ is valid.

3. $(A \rightarrow B \wedge B \rightarrow C) \rightarrow (A \rightarrow C)$

Assume $(A \rightarrow B \wedge B \rightarrow C) \rightarrow (A \rightarrow C)$ is not valid
 $\Rightarrow$ There is some v such that $v \not\models (A \rightarrow B \wedge B \rightarrow C) \rightarrow (A \rightarrow C)$
 $\Rightarrow v \models (A \rightarrow B \wedge B \rightarrow C)$ and $v \not\models A \rightarrow C$
 $\Rightarrow v \models A \rightarrow B$ and $v \models B \rightarrow C$
 $\Rightarrow v \not\models A$ or $v \models A \wedge B$
 $\Rightarrow v \models A \rightarrow C$ vacuously or $v \models B$ and $v \models A$
 $\Rightarrow v \models C$ as $v \models B \rightarrow C$ and $v \models B$
 $\Rightarrow v \models A$ and $v \models C$
 $\Rightarrow v \models A \rightarrow C$

Therefore, by contradiction in either case of $v \models A \rightarrow B$, $(A \rightarrow B \wedge B \rightarrow C) \rightarrow (A \rightarrow C)$ is valid.

4. $(A \rightarrow B) \leftrightarrow (\neg B \rightarrow \neg A)$

Before starting this, lets come up with some simplifications of implications and bi-conditionals.

$$v \models P \rightarrow Q \iff v \not\models P \text{ or } v \models P \wedge Q \iff v \models \neg P \vee (P \wedge Q) \iff v \models \neg P \vee Q$$

$$v \not\models P \rightarrow Q \iff v \not\models \neg P \vee Q \iff v \models \neg(\neg P \vee Q) \iff v \models P \wedge \neg Q$$

Now for bi-conditional:

$$\begin{aligned}
 v \models P \leftrightarrow Q &\iff v \models (P \rightarrow Q) \wedge (Q \rightarrow P) \\
 &\iff v \models (\neg P \vee Q) \wedge (\neg Q \vee P)
 \end{aligned}$$

$$\begin{aligned}
 v \not\models P \leftrightarrow Q &\iff v \not\models (P \rightarrow Q) \wedge (Q \rightarrow P) \\
 &\iff v \models \neg((P \rightarrow Q) \wedge (Q \rightarrow P)) \\
 &\iff v \models \neg(P \rightarrow Q) \vee \neg(Q \rightarrow P) \\
 &\iff v \models (P \wedge \neg Q) \vee (Q \wedge \neg P)
 \end{aligned}$$

Assume $(A \rightarrow B) \leftrightarrow (\neg B \rightarrow \neg A)$ is not valid

$$\begin{aligned}
 \not\models (A \rightarrow B) \leftrightarrow (\neg B \rightarrow \neg A) &\iff \text{for some } v, v \not\models (A \rightarrow B) \leftrightarrow (\neg B \rightarrow \neg A) \\
 &\iff v \models (A \rightarrow B \wedge \neg(\neg B \rightarrow \neg A)) \vee ((\neg B \rightarrow \neg A) \wedge \neg(A \rightarrow B))
 \end{aligned}$$

$$\begin{aligned}
 \text{Left Branch } v \models (A \rightarrow B \wedge \neg(\neg B \rightarrow \neg A)) & \\
 \iff v \models ((\neg A \vee B) \wedge \neg(B \vee \neg A)) & \\
 \iff v \models (\neg A \vee B) \wedge (\neg B \wedge A) & \\
 \iff v(B) = F \text{ and } v(A) = T \text{ by } v \models (\neg B \wedge A) & \\
 \iff \text{but } v \models \neg A \vee B \text{ requires } v(A) = F \text{ or } v(B) = T & \\
 \iff \text{which is a contradiction in either case} &
 \end{aligned}$$

$$\begin{aligned}
 \text{Right Branch } v \models ((\neg B \rightarrow \neg A) \wedge \neg(A \rightarrow B)) & \\
 \iff v \models (B \vee \neg A) \wedge \neg(\neg A \vee B) & \\
 \iff v \models (B \vee \neg A) \wedge (A \wedge \neg B) & \\
 \iff v(B) = F \text{ and } v(A) = T \text{ by } v \models (A \wedge \neg B) & \\
 \iff \text{but } v \models B \vee \neg A \text{ requires } v(A) = F \text{ or } v(B) = T &
 \end{aligned}$$

Therefore, $(A \rightarrow B) \leftrightarrow (\neg B \rightarrow \neg A)$ is valid.

4. Prove the following:

1. $\models \varphi \iff \neg\varphi$ is not satisfiable
 2. PROP-VAL reduces to PROP-SAT
 3. PROP-SAT reduces to PROP-VAL
-

1. $\models \varphi \iff \neg\varphi$ is not satisfiable

Let $\models \varphi$

$\iff$ For all $v, v \models \varphi$

$\iff$ For all $v, v \not\models \neg\varphi$

$\iff \neg\varphi$ is not satisfiable

2. PROP-VAL reduces to PROP-SAT

Let M_{PS} be a decider for PROP-SAT.

define M_{PV} on $\langle\varphi\rangle$
 modify $\langle\varphi\rangle$ into $\langle\neg\varphi\rangle$
 return not $M_{PS}(\langle\neg\varphi\rangle)$

Modifying $\langle\varphi\rangle$ into $\langle\neg\varphi\rangle$ is computable as it is inserting a single negation into the formula representation at its root. The not statement is decidable, and M_{PS} is a decider. Therefore M_{PV} is a decider.

$$\begin{aligned} M_{PV}(\langle\varphi\rangle) = 1 &\iff M_{PS}(\langle\neg\varphi\rangle) = 0 \\ &\iff \langle\neg\varphi\rangle \notin \text{PROP-SAT} \\ &\iff \neg\varphi \text{ is not satisfiable} \\ &\iff \models \varphi \\ &\iff \langle\varphi\rangle \in \text{PROP-VAL} \end{aligned}$$

Therefore PROP-VAL reduces to PROP-SAT.

3. PROP-SAT reduces to PROP-VAL

Let M_{PV} be a decider for PROP-VAL.

define M_{PS} on $\langle\varphi\rangle$
 modify $\langle\varphi\rangle$ into $\langle\neg\varphi\rangle$
 return not $M_{PV}(\langle\neg\varphi\rangle)$

Modifying $\langle\varphi\rangle$ into $\langle\neg\varphi\rangle$ is computable as we adding a single negation into the representation. A not statement is decidable, and M_{PV} is a decider. Therefore M_{PS} is a decider.

$$\begin{aligned}
M_{\text{PS}}(\langle \varphi \rangle) = 1 &\iff M_{\text{PV}}(\langle \neg \varphi \rangle) = 0 \\
&\iff \not\models \neg \varphi \\
&\iff \text{There is some } v \text{ such that } v \not\models \neg \varphi \\
&\iff v \models \varphi \\
&\iff \text{There is some } v \text{ such that } v \models \varphi \\
&\iff \varphi \text{ is satisfiable} \\
&\iff \langle \varphi \rangle \in \text{PROP-SAT}
\end{aligned}$$

Therefore PROP-SAT reduces to PROP-VAL

5. Prove (or disprove) the following formula are a logical consequence of the finite sets:

- You may *not* assume that $\Lambda \models \varphi \iff \models \bigwedge \Lambda \rightarrow \varphi$

- $\{A, A \rightarrow B\} \models B$
 - $\{\neg B, A \rightarrow B\} \models \neg A$
 - $\{A \rightarrow B, B \rightarrow C\} \models A \rightarrow C$
-

- $\{A, A \rightarrow B\} \models B$

$$\begin{aligned}
&\text{Let } v \models A \wedge (A \rightarrow B) \\
&\Rightarrow v \models A \wedge B \\
&\Rightarrow v \models B
\end{aligned}$$

Therefore $\{A, A \rightarrow B\} \models B$ as for any v such that $v \models A \wedge (A \rightarrow B)$ then $v \models B$

- $\{\neg B, A \rightarrow B\} \models \neg A$

$$\begin{aligned}
&\text{Let } v \models \neg B \wedge A \rightarrow B \\
&\Rightarrow v \not\models A \text{ or } v \models A \wedge B \text{ but } v \not\models B \\
&\Rightarrow v \not\models A \Rightarrow v \models \neg A
\end{aligned}$$

Therefore $\{\neg B, A \rightarrow B\} \models \neg A$

- $\{A \rightarrow B, B \rightarrow C\} \models A \rightarrow C$

$$\begin{aligned}
&\text{Let } v \models (A \rightarrow B) \wedge (B \rightarrow C) \\
&\Rightarrow v \models A \rightarrow B \text{ and } v \models B \rightarrow C \\
&\Rightarrow v \not\models A \text{ or } v \models A \wedge B
\end{aligned}$$

$$\begin{aligned}
&\text{Case I : } v \not\models A \\
&\Rightarrow v \models A \rightarrow C \text{ vacuously}
\end{aligned}$$

$$\begin{aligned}
&\text{Case II : } v \models A \wedge B \text{ and } v \models B \rightarrow C \\
&\Rightarrow v \models B \text{ and } v \models B \rightarrow C \\
&\Rightarrow v \models C \\
&\Rightarrow v \models A \wedge C \\
&\Rightarrow v \models A \rightarrow C
\end{aligned}$$

Therefore, in any case, $\{A \rightarrow B, B \rightarrow C\} \models A \rightarrow C$

6. Prove the following:

1. $\Lambda \models \varphi \iff \models \bigwedge \Lambda \rightarrow \varphi$
 2. $\text{PROP-VAL} \leq_m \text{PROP-CONQ}$
 3. $\text{PROP-CONQ} \leq_m \text{PROP-VAL}$
-

1. $\Lambda \models \varphi \iff \models \bigwedge \Lambda \rightarrow \varphi$

Let $\Lambda \models \varphi$

Case I: Let $v \models \bigwedge \Lambda$

$$\iff v \models \varphi$$

$$\iff v \models \bigwedge \Lambda \wedge v \models \varphi$$

$$\iff v \models \bigwedge \Lambda \rightarrow \varphi$$

Case II: Let $v \not\models \bigwedge \Lambda$

$$\iff v \models \bigwedge \Lambda \rightarrow \varphi \text{ vacuously}$$

Therefore by case I and II, $\Lambda \models \varphi \iff \models \bigwedge \Lambda \rightarrow \varphi$

2. $\text{PROP-VAL} \leq_m \text{PROP-CONQ}$

Let $f(\langle \Lambda, \varphi \rangle) = \langle \bigwedge \Lambda \rightarrow \varphi \rangle$

$$\langle \Lambda, \varphi \rangle \in \text{PROP-CONQ} \iff \Lambda \models \varphi$$

$$\iff \models \bigwedge \Lambda \rightarrow \varphi$$

$$\iff \langle \bigwedge \Lambda \rightarrow \varphi \rangle \in \text{PROP-VAL}$$

$$\iff f(\langle \Lambda, \varphi \rangle) \in \text{PROP-VAL}$$

3. $\text{PROP-CONQ} \leq_m \text{PROP-VAL}$

This requires a bit more effort.

We need to convert a valid formula into the form of a consequence statement. Validity means that φ is always true, or in the language of consequence, if nothing is true then φ is true.

So we first need to prove that

$$\models \varphi \iff \emptyset \models \varphi$$

We need to extend our notation a little bit: What are the semantics of v and $\emptyset$?

This is similar to the idea of vacuous truth in implication. There, we claim that an implication with a false antecedent is 'true' because we cannot *disprove* it. There is no way to show that if the sky was green, then grass is green, because the hypothetical of a green sky is false. We cannot find some green sky with blue grass to be a counterexample for this claim.

For a 'nothing' formula, it is always true, because there is nothing to disprove. So, for every v , $v \models \emptyset$.

$$\text{RTP: } \models \varphi \iff \emptyset \models \varphi$$

Let $\models \varphi$
 $\Rightarrow$ for every $v, v \models \varphi$
 $v \models \emptyset$ as well
 $\Rightarrow v \models \emptyset \wedge \varphi$
 $\Rightarrow v \models \emptyset \rightarrow \varphi$
 $\Rightarrow \emptyset \models \varphi$

Let $\emptyset \models \varphi$
 $\Rightarrow \models \emptyset \rightarrow \varphi$
 $\Rightarrow$ for every $v, v \models \emptyset \rightarrow \varphi$
but $v \models \emptyset$ for every v
 $\Rightarrow v \models \varphi$
 $\Rightarrow \models \varphi$

Let $f(\langle \varphi \rangle) = \langle \emptyset, \varphi \rangle$

$\langle \varphi \rangle \in \text{PROP-VAL} \iff \models \varphi$
 $\iff \emptyset \models \varphi$
 $\iff \langle \emptyset, \varphi \rangle \in \text{PROP-CONQ}$
 $\iff f(\langle \varphi \rangle) \in \text{PROP-CONQ}$

7. Prove that PROP-CONQ reduces to PROP-SAT

We know the following from previous questions:

1. $\models \varphi \iff \neg \varphi$ is not satisfiable
2. $\models \bigwedge \Lambda \rightarrow \varphi \iff \Lambda \models \varphi$

Therefore $\neg(\bigwedge \Lambda \rightarrow \varphi)$ is not satisfiable $\iff \Lambda \models \varphi$

Using this, we can put together the final reduction.

Assume that PROP-SAT is decidable, then M_{PS} decides it.

define M_{PC} on $\langle \Lambda, \varphi \rangle$
 modify $\langle \Lambda, \varphi \rangle$ into $\langle \neg(\bigwedge \Lambda \rightarrow \varphi) \rangle$
 return not $M_{\text{PS}}(\langle \neg(\bigwedge \Lambda \rightarrow \varphi) \rangle)$

The string modification is computable, not is decidable and M_{PS} is a decider. Therefore M_{PC} is a decider.

$M_{\text{PC}}(\langle \Lambda, \varphi \rangle) = 1 \iff M_{\text{PS}}(\langle \neg(\bigwedge \Lambda \rightarrow \varphi) \rangle) = 0$
 $\iff \langle \neg(\bigwedge \Lambda \rightarrow \varphi) \rangle \notin \text{PROP-SAT}$
 $\iff \neg(\bigwedge \Lambda \rightarrow \varphi)$ is not satisfiable
 $\iff \Lambda \models \varphi$
 $\iff \langle \Lambda, \varphi \rangle \in \text{PROP-CONQ}$

Hence M_{PC} decides PROP-CONQ using the decider M_{PS} for PROP-SAT. Therefore PROP-CONQ reduces to PROP-SAT
